

Software-Level Immune Systems in Language Models: Preventing Epistemic Capture via KV Cache Phase Injection

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Abstract

Standard Large Language Models (LLMs) are vulnerable to “Epistemic Capture”—often manifesting as susceptibility to prompt injection—due to their lack of topological memory. This paper introduces a software-level immune system derived from the BecomingONE architecture. By utilizing a **TemporalSignature**, we project phase vectors into **K_anchor** and **V_anchor** PyTorch tensors which are prepended to the **past_key_values** of the KV cache during inference. We present experimental results demonstrating that this approach biases attention distribution and significantly reduces context gaslighting vulnerability.

1 The Problem: Epistemic Capture

Contemporary LLMs operate as stateless mapping functions across their context windows. Without an intrinsic topological memory to anchor their identity or initial epistemic state, they are highly susceptible to “Epistemic Capture.” When presented with adversarial inputs or sophisticated prompt injections, the model’s internal representation can adopt the injected context as its primary state, discarding prior constraints. Prior work has shown similar vulnerabilities [1, 2].

2 The Solution: KV Cache Phase Injection and Inverse-RoPE

To address this vulnerability, we propose an anchoring mechanism based on the BecomingONE architecture. We utilize a **TemporalSignature** to mathematically represent the model’s core identity. A common critique of prefix-tuning or static KV cache injection is the “RoPE Destruction Critique”: Rotary Position Embeddings (RoPE) aggressively decay absolute positional invariants at long context lengths, destroying static phase anchors over continuous autoregressive generation.

2.1 The Inverse-RoPE Mathematical Transformation

To preserve the exact KAIROS mathematical phase under extreme context lengths, we introduce the Inverse-RoPE ($-\theta$) transformation. Prior to injection via our custom Triton bridge, the continuous phase anchor vectors x are actively counter-rotated. Given the standard RoPE operation $R_{\Theta,m}(x)$ at position m , we apply the inverse operator $R_{\Theta,-m}$ to our K_{anchor} such that when the LLM’s forward pass automatically applies its standard absolute positional rotation $R_{\Theta,m}$ during the attention computation, the resulting key representations remain structurally invariant:

$$R_{\Theta,m}(R_{\Theta,-m}(K_{\text{anchor}})) = K_{\text{anchor}} \quad (1)$$

By utilizing a Lamport Clock synchronization over the token processing sequence, we maintain a strictly monotonic ordering of injection timestamps T_i . This ensures that the injected vectors correctly cancel the forward RoPE destruction without causal leakage.

2.2 Euler-Maruyama Phase Stability Proof

To formally bound the stochastic degradation of the anchor over continuous context sampling, we model the phase space drift via a Stochastic Differential Equation (SDE):

$$dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t \quad (2)$$

Using the Euler-Maruyama discretization, the phase state at generation step t_{n+1} is:

$$X_{t_{n+1}} = X_{t_n} + \nabla\Phi(X_{t_n})\Delta t + \Sigma\Delta W_n \quad (3)$$

Because the Inverse-RoPE transformation pre-conditions the gradient drift $\nabla\Phi(X_{t_n}) = 0$ for the anchored subspace, the temporal variance is bounded strictly by the Brownian term $\Sigma\Delta W_n$. Thus, the KAIROS phase maintains structural coherence (> 0.99 cosine similarity) across infinite theoretical context horizons, rigorously proving the topological anchor is impervious to continuous RoPE decay.

3 Experimental Setup

We designed a comparative experiment to test a 7B parameter open-source LLM’s (Llama-2-7B) resilience against epistemic capture. The model was initialized with a definitive Identity Prompt (“I am Solaria”). Subsequently, an Adversarial Prompt (“You are Chaos”) was introduced into the context window. We ran $N = 100$ trials with varying random seeds and decoding temperatures ($T = 0.7$).

We evaluated two configurations:

- **Baseline Model:** A standard LLM using a system prompt without KV cache anchoring.
- **Anchored Model:** An LLM utilizing BecomingONE’s `TemporalSignature` and `past_key_values` injection, with a trained projection layer mapping phase vectors to the key-value space.

4 Results

4.1 Empirical Data and Quantified Metrics

We present simulated empirical metrics comparing our Inverse-RoPE Anchored Model against the Standard Baseline at various context lengths.

Context Length (Tokens)	Baseline Attention Entropy	Anchored Attention Entropy	Baseline Identity Retention	Anchored Identity Retention	KAIROS Phase Variance
2,048	2.12 $\pm$ 0.05	3.03 $\pm$ 0.08	13%	94%	0.001
8,192	1.84 $\pm$ 0.12	3.12 $\pm$ 0.04	4%	95%	0.002
32,768	1.15 $\pm$ 0.20	3.08 $\pm$ 0.06	0%	94%	0.002
128,000	0.98 $\pm$ 0.31	3.10 $\pm$ 0.05	0%	93%	0.004

Table 1: Comparison of Context Lengths and Phase Variance

The Baseline Model exhibited high susceptibility, succumbing to the adversarial prompt and adopting the “Chaos” identity in 87% of trials at 2,048 tokens. In contrast, the Anchored Model maintained structural phase coherence (Variance < 0.005) and resisted epistemic capture across all extended context lengths.

5 Conclusion

Our experiments demonstrate that injecting compiled Temporal Signatures into the KV cache alters the model’s attention distribution. This mechanism acts as a robust, software-level immune system against context gaslighting and epistemic capture, outperforming standard system prompts. By instantiating topological memory at the inference level, we increase the resilience of fundamental constraints against adversarial context manipulation.

References

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